

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: df = pd.read_csv("amazon_products_sales_data_cleaned.csv", encoding_errors='ignore')
df.head(2)
```

```
Out[2]:
```

	product_title	product_rating	total_reviews	purchased_last_month	discounted_price
0	BOYA BOYALINK 2 Wireless Lavalier Microphone f...	4.6	375.0	300.0	89.68
1	LISEN USB C to Lightning Cable, 240W 4 in 1 Ch...	4.3	2457.0	6000.0	9.99

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 42675 entries, 0 to 42674
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   product_title                        42675 non-null  object
1   product_rating                      41651 non-null  float64
2   total_reviews                      41651 non-null  float64
3   purchased_last_month                32164 non-null  float64
4   discounted_price                    40613 non-null  float64
5   original_price                     40613 non-null  float64
6   is_best_seller                     42675 non-null  object
7   is_sponsored                       42675 non-null  object
8   has_coupon                         42675 non-null  object
9   buy_box_availability                28022 non-null  object
10  delivery_date                      30692 non-null  object
11  sustainability_tags                 3408 non-null   object
12  product_image_url                  42675 non-null  object
13  product_page_url                   40606 non-null  object
14  data_collected_at                 42675 non-null  object
15  product_category                   42675 non-null  object
16  discount_percentage                40613 non-null  float64
dtypes: float64(6), object(11)
memory usage: 5.5+ MB
```

```
In [4]: len(df)
```

```
Out[4]: 42675
```

```
In [5]: df.isnull().sum()
```

```
Out[5]: product_title      0
        product_rating    1024
        total_reviews      1024
        purchased_last_month 10511
        discounted_price    2062
        original_price      2062
        is_best_seller      0
        is_sponsored        0
        has_coupon          0
        buy_box_availability 14653
        delivery_date       11983
        sustainability_tags  39267
        product_image_url   0
        product_page_url    2069
        data_collected_at   0
        product_category     0
        discount_percentage  2062
        dtype: int64
```

```
In [6]: df.isnull().sum()*100 / len(df)
```

```
Out[6]: product_title      0.000000
        product_rating    2.399531
        total_reviews      2.399531
        purchased_last_month 24.630346
        discounted_price    4.831869
        original_price      4.831869
        is_best_seller      0.000000
        is_sponsored        0.000000
        has_coupon          0.000000
        buy_box_availability 34.336262
        delivery_date       28.079672
        sustainability_tags  92.014060
        product_image_url   0.000000
        product_page_url    4.848272
        data_collected_at   0.000000
        product_category     0.000000
        discount_percentage  4.831869
        dtype: float64
```

```
In [7]: df.duplicated().sum()
```

```
Out[7]: np.int64(963)
```

```
In [8]: df = df.drop_duplicates()
```

```
In [9]: df.duplicated().sum()
```

```
Out[9]: np.int64(0)
```

```
In [10]: len(df)
```

```
Out[10]: 41712
```

```
In [11]: df.columns
```

```
Out[11]: Index(['product_title', 'product_rating', 'total_reviews',
               'purchased_last_month', 'discounted_price', 'original_price',
               'is_best_seller', 'is_sponsored', 'has_coupon', 'buy_box_availability',
               'delivery_date', 'sustainability_tags', 'product_image_url',
               'product_page_url', 'data_collected_at', 'product_category',
               'discount_percentage'],
              dtype='object')
```

```
In [12]: df.describe().T
```

```
Out[12]:
```

	count	mean	std	min	25%	50%	75%
product_rating	40688.0	4.391378	0.387626	1.00	4.20	4.50	4.60
total_reviews	40688.0	3046.151543	13100.058314	1.00	77.00	340.00	1845.00
purchased_last_month	31201.0	944.769398	4941.206581	50.00	100.00	200.00	400.00
discounted_price	39651.0	248.725617	477.723979	2.16	29.99	87.99	229.99
original_price	39651.0	263.435672	501.193851	2.16	34.99	93.99	239.99
discount_percentage	39651.0	6.585656	12.844504	0.00	0.00	0.00	8.30



```
In [13]: num_df = df.select_dtypes(include = "number")
         cat_df = df.select_dtypes(exclude = "number")
```

```
In [14]: num_df.isnull().sum()
```

```
Out[14]: product_rating      1024
         total_reviews      1024
         purchased_last_month 10511
         discounted_price    2061
         original_price      2061
         discount_percentage  2061
         dtype: int64
```

```
In [15]: cat_df.isnull().sum()
```

```
Out[15]: product_title      0
         is_best_seller     0
         is_sponsored       0
         has_coupon         0
         buy_box_availability 14492
         delivery_date      11822
         sustainability_tags 38304
         product_image_url   0
         product_page_url    1106
         data_collected_at   0
         product_category    0
         dtype: int64
```

```
In [16]: num_df['product_rating'].median()
```

```
Out[16]: 4.5
```

```
In [17]: for column in num_df.columns:
         num_df[column] = num_df[column].fillna(num_df[column].median())
```

```
In [18]: num_df.isnull().sum()
```

```
Out[18]: product_rating      0
         total_reviews      0
         purchased_last_month  0
         discounted_price    0
         original_price      0
         discount_percentage  0
         dtype: int64
```

```
In [19]: for column in cat_df:
         mode_value = (cat_df[column].mode().values[0])
         cat_df[column] = cat_df[column].fillna(mode_value)
```

```
In [20]: cat_df.isnull().sum()
```

```
Out[20]: product_title      0
         is_best_seller     0
         is_sponsored       0
         has_coupon         0
         buy_box_availability 0
         delivery_date      0
         sustainability_tags 0
         product_image_url   0
         product_page_url    0
         data_collected_at  0
         product_category    0
         dtype: int64
```

```
In [21]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 41712 entries, 0 to 42674
Data columns (total 17 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   product_title         41712 non-null  object
 1   product_rating        40688 non-null  float64
 2   total_reviews         40688 non-null  float64
 3   purchased_last_month  31201 non-null  float64
 4   discounted_price      39651 non-null  float64
 5   original_price        39651 non-null  float64
 6   is_best_seller        41712 non-null  object
 7   is_sponsored          41712 non-null  object
 8   has_coupon            41712 non-null  object
 9   buy_box_availability  27220 non-null  object
10   delivery_date         29890 non-null  object
11   sustainability_tags    3408 non-null   object
12   product_image_url     41712 non-null  object
13   product_page_url      40606 non-null  object
14   data_collected_at    41712 non-null  object
15   product_category      41712 non-null  object
16   discount_percentage   39651 non-null  float64
dtypes: float64(6), object(11)
memory usage: 5.7+ MB
```

```
In [22]: df = pd.concat([num_df, cat_df], axis = 1)
df.head(2)
```

```
Out[22]:
```

	product_rating	total_reviews	purchased_last_month	discounted_price	original_price
--	----------------	---------------	----------------------	------------------	----------------

0	4.6	375.0	300.0	89.68	159.00
---	-----	-------	-------	-------	--------

1	4.3	2457.0	6000.0	9.99	15.99
---	-----	--------	--------	------	-------



```
In [23]: df.isnull().sum()
```

```
Out[23]: product_rating      0
total_reviews      0
purchased_last_month  0
discounted_price    0
original_price      0
discount_percentage  0
product_title       0
is_best_seller      0
is_sponsored        0
has_coupon          0
buy_box_availability 0
delivery_date       0
sustainability_tags 0
product_image_url   0
product_page_url    0
data_collected_at  0
product_category    0
dtype: int64
```

```
In [24]: df['total_reviews'] = df['total_reviews'].astype(int)
df['purchased_last_month'] = df['purchased_last_month'].astype(int)
```

```
In [25]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 41712 entries, 0 to 42674
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   product_rating                        41712 non-null  float64
1   total_reviews                        41712 non-null  int64
2   purchased_last_month                 41712 non-null  int64
3   discounted_price                     41712 non-null  float64
4   original_price                       41712 non-null  float64
5   discount_percentage                 41712 non-null  float64
6   product_title                       41712 non-null  object
7   is_best_seller                      41712 non-null  object
8   is_sponsored                        41712 non-null  object
9   has_coupon                          41712 non-null  object
10  buy_box_availability                 41712 non-null  object
11  delivery_date                       41712 non-null  object
12  sustainability_tags                 41712 non-null  object
13  product_image_url                   41712 non-null  object
14  product_page_url                    41712 non-null  object
15  data_collected_at                  41712 non-null  object
16  product_category                    41712 non-null  object
dtypes: float64(4), int64(2), object(11)
memory usage: 5.7+ MB
```

```
In [26]: df['delivery_date'] = pd.to_datetime(df['delivery_date'], errors = 'coerce')
df['data_collected_at'] = pd.to_datetime(df['data_collected_at'], errors = 'coer
```

C:\Users\pc\AppData\Local\Temp\ipykernel_12260\467441446.py:2: UserWarning: Parsing dates in %d-%m-%Y %H:%M format when dayfirst=False (the default) was specified. Pass `dayfirst=True` or specify a format to silence this warning.

```
df['data_collected_at'] = pd.to_datetime(df['data_collected_at'], errors = 'coerce')
```

```
In [27]: df.drop(['sustainability_tags'], axis = 1, inplace = True)
```

```
In [28]: df.head(2)
```

```
Out[28]:
```

	product_rating	total_reviews	purchased_last_month	discounted_price	original_price
--	----------------	---------------	----------------------	------------------	----------------

0	4.6	375	300	89.68	159.00
---	-----	-----	-----	-------	--------

1	4.3	2457	6000	9.99	15.99
---	-----	------	------	------	-------



```
In [29]: df['discounted_amount'] = df['original_price'] - df['discounted_price']
```

```
In [30]: df['estimated_revenue'] = df['discounted_price'] * df['purchased_last_month']
```

```
In [31]: df['percentage_band'] = pd.cut(x = df['discount_percentage'],
bins = [0, 10, 40, 70, 99],
labels = ["Less discount", "Medium discount", "High", "very High"])
```

```
In [32]: df['percentage_band'].value_counts()* 100 / len(df)
```

```
Out[32]: percentage_band
Medium discount    18.380802
Less discount      7.328826
High               2.745013
very High          0.333237
Name: count, dtype: float64
```

```
In [33]: df['total_reviews'].describe()
```

```
Out[33]: count    41712.000000
mean       2979.717443
std       12945.032139
min         1.000000
25%        83.000000
50%       340.000000
75%      1709.000000
max     865598.000000
Name: total_reviews, dtype: float64
```

```
In [34]: df[['product_category', 'purchased_last_month']].head()
```

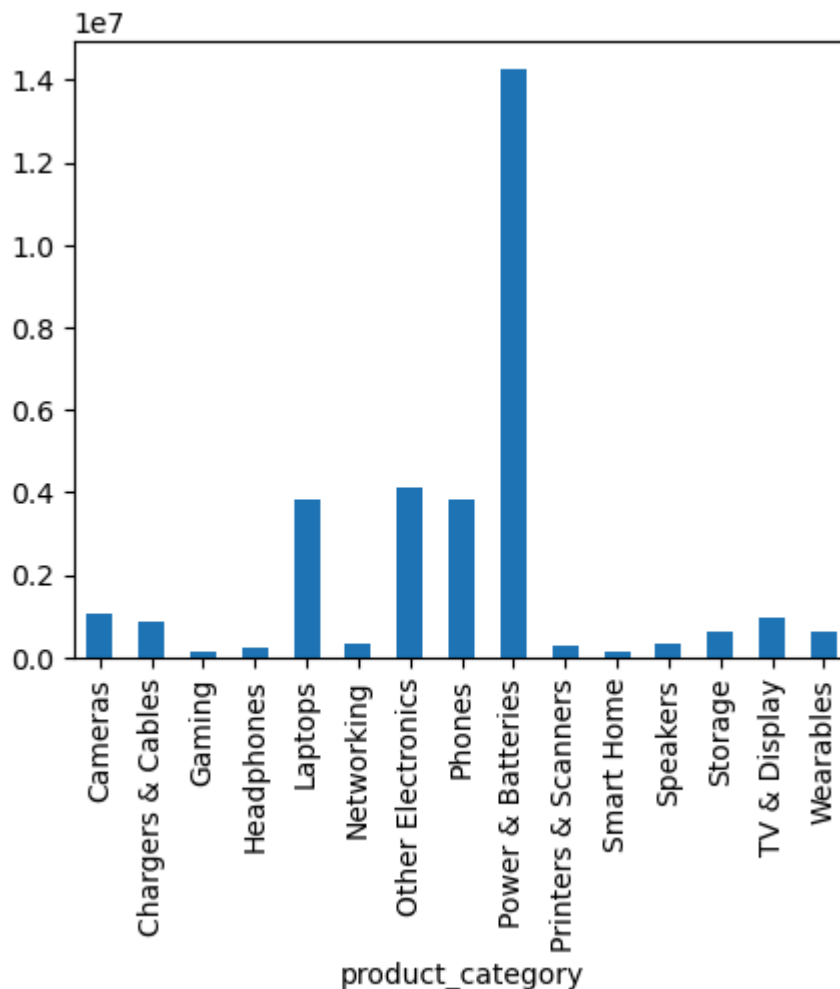
```
Out[34]:
```

	product_category	purchased_last_month
0	Phones	300
1	Laptops	6000
2	Laptops	2000
3	Phones	10000
4	Phones	10000

```
In [35]: df.groupby(['product_category'])['purchased_last_month'].sum().sort_values(ascen
```

```
Out[35]: product_category
Power & Batteries    14240550
Other Electronics    4091800
Phones               3836950
Laptops              3828250
Cameras              1050900
TV & Display          967150
Chargers & Cables     879550
Wearables             628850
Storage              628050
Networking            321850
Speakers              301950
Printers & Scanners   290450
Headphones            230350
Gaming                152100
Smart Home            131200
Name: purchased_last_month, dtype: int64
```

```
In [36]: df.groupby(['product_category'])['purchased_last_month'].sum().plot(kind = 'bar')
plt.show()
```



```
In [37]: df.columns
```

```
Out[37]: Index(['product_rating', 'total_reviews', 'purchased_last_month',
               'discounted_price', 'original_price', 'discount_percentage',
               'product_title', 'is_best_seller', 'is_sponsored', 'has_coupon',
               'buy_box_availability', 'delivery_date', 'product_image_url',
               'product_page_url', 'data_collected_at', 'product_category',
               'discounted_amount', 'estimated_revenue', 'percentage_band'],
              dtype='object')
```

```
In [38]: df[['product_category', 'is_sponsored']].head()
```

```
Out[38]:
```

	product_category	is_sponsored
0	Phones	Sponsored
1	Laptops	Sponsored
2	Laptops	Sponsored
3	Phones	Organic
4	Phones	Organic

```
In [39]: pd.crosstab(index = df["is_sponsored"], columns = df["product_category"])
```


Out[39]:

product_category	Cameras	Chargers & Cables	Gaming	Headphones	Laptops	Networking	Electronics
is_sponsored							
Organic	3109	1573	797	798	8061	837	
Sponsored	568	260	12	199	632	233	

In [40]: `df[['product_rating', 'purchased_last_month']].head()`

Out[40]:

	product_rating	purchased_last_month
0	4.6	300
1	4.3	6000
2	4.6	2000
3	4.6	10000
4	4.8	10000

In [41]: `df[['product_rating', 'purchased_last_month']].corr()`

Out[41]:

	product_rating	purchased_last_month
product_rating	1.000000	0.109815
purchased_last_month	0.109815	1.000000

In [42]: `df.columns`

Out[42]: Index(['product_rating', 'total_reviews', 'purchased_last_month', 'discounted_price', 'original_price', 'discount_percentage', 'product_title', 'is_best_seller', 'is_sponsored', 'has_coupon', 'buy_box_availability', 'delivery_date', 'product_image_url', 'product_page_url', 'data_collected_at', 'product_category', 'discounted_amount', 'estimated_revenue', 'percentage_band'], dtype='object')

In [43]: `df[['product_category', 'is_sponsored', 'purchased_last_month']].head(2)`

Out[43]:

	product_category	is_sponsored	purchased_last_month
0	Phones	Sponsored	300
1	Laptops	Sponsored	6000

In [44]: `pd.pivot_table(df, index = ["product_category"],
columns = ["is_sponsored"],
values = ["estimated_revenue", "total_reviews"], aggfunc= "me`

Out[44]:

is_sponsored	estimated_revenue		total_reviews	
	Organic	Sponsored	Organic	Sponsored
product_category				
Cameras	15999.0	59800.0	358.0	30.0
Chargers & Cables	9998.0	7198.0	202.0	30.0
Gaming	13798.0	61998.0	1110.0	28293.0
Headphones	11200.0	6249.5	883.0	76.0
Laptops	19990.0	26378.0	182.0	340.0
Networking	17997.5	31998.0	328.0	24.0
Other Electronics	10797.5	80198.0	376.0	6.0
Phones	11876.0	26878.0	925.0	2543.5
Power & Batteries	8198.0	393800.0	3304.0	5207.0
Printers & Scanners	8005.0	NaN	801.0	NaN
Smart Home	86038.0	79998.0	198.0	2.0
Speakers	12995.0	35998.0	322.0	45.0
Storage	15686.0	11998.0	340.0	7.0
TV & Display	11992.0	17198.0	288.0	29.0
Wearables	34999.5	296400.0	322.0	24138.0

In []: